

Measuring ℓ_∞ Attacks by the ℓ_2 Norm

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Abstract

Deep Neural Networks (DNNs) could be easily fooled by Adversarial Examples (AEs) with the imperceptible difference to original samples in human eyes. To keep the difference imperceptible, the existing attacking bound the adversarial perturbations by the ℓ_∞ norm, which is then served as the standard to align different attacks for a fair comparison. However, when investigating attack transferability, i.e., the capability of the AEs from attacking one surrogate DNN to cheat other black-box DNN, we find that

only using the ℓ_∞ norm is not sufficient to measure the attack strength, according to our comprehensive experiments concerning 7 transfer-based attacks, 4 white-box surrogate models, and 9 black-box victim models. Specifically, we find that the ℓ_2 norm greatly affects the transferability in ℓ_∞ attacks. Since larger-perturbed AEs naturally bring about better transferability, we advocate that the strength of all attacks should be measured by both the widely used ℓ_∞ and also the ℓ_2 norm. Despite the intuitiveness of our conclusion and advocacy, they are very necessary for the community, because common evaluations (bounding only the ℓ_∞ norm) allow tricky enhancements of the “attack transferability” by increasing the “attack strength” (ℓ_2 norm) as shown by our simple counter-example method, and the good transferability of several existing methods may be due to their large ℓ_2 distances.

1 Introduction

Deep Learning (DL) has widely served as a mainstream tool in various fields, and its vulnerability has also attracted lots of attention in recent years, among which an apparent instance is the existence of Adversarial Examples (AEs) (Goodfellow et al., 2015; Carlini and Wagner, 2017; Madry et al., 2018; Tang et al., 2021). AEs are slightly perturbed from the natural samples but could cheat *victim* Deep Neural Networks (DNNs) to produce incorrect predictions (Szegedy et al., 2014). It is also found that white-box AEs on a *surrogate* model have transferability to cheat other unknown victim models, i.e., the *transfer-based attacks* (Dong et al., 2018; Xie et al., 2019b).

In adversarial attacks, fair evaluations matter, and proper measurement of the at-

attack strength is fundamental for distinguishing between the performance improvements from the settings (strength) and from the methods (transferability). The latter reveals underlying mechanisms of adversarial attacks, while the former simply bypasses the (inadequate) rules for good numerical results. Currently, almost all transfer-based attacks bound the AEs by the ℓ_∞ ball with a fixed radius (Dong et al., 2018; Xie et al., 2019b; Dong et al., 2019; Lin et al., 2020; Chen et al., 2021a; Wu et al., 2019; Guo et al., 2020; Wan et al., 2021; Huang et al., 2021; Xie et al., 2021; Gao et al., 2020), assuming that on this occasion, the strength of attacks is maintained the same for a fair evaluation of the attack transferability. However, it is not the case considering an extreme example: an attacker changes only one pixel (Su et al., 2019), while another attacker modifies all pixels to the ℓ_∞ bound (Goodfellow et al., 2015). Although their perturbations are the same under the measurement of ℓ_∞ norm, their attack strength is obviously different.

In this paper, we reveal the insufficient measurement of the attack strength by demonstrating that the ℓ_2 distances greatly affect the transferability of AEs under the same ℓ_∞ bound, indicating that the ℓ_2 norm is also a non-negligible measure. We investigate 7 excellent transfer-based attacks, which lead to very different ℓ_2 distances for our study. We perform comprehensive experiments concerning 4 surrogates and 9 victims (including adversarially-trained ones) with diverse architectures and find three phenomena that support our claim.

- Attacks that produce AEs with larger ℓ_2 distances from original samples result in higher-transferable AEs.
- Increasing the ℓ_2 distances of AEs leads to a distinctive boost in transferability

when the attack is conducted in an unchanged manner.

- The transferability of AEs from almost all attacks stays amazingly the same when the ℓ_2 distances is controlled to a similar value.

Given these observations, we see that the ℓ_2 norm influences the transferability a lot, and thus is an important but long-neglected measurement of the attack strength in ℓ_∞ attacks. To further illustrate that, we offer a simple counter-example to show the possibility to seemingly “enhance” the transferability by actually just increasing the attack strength, and point out that the good transferability of several existing methods may be due to the large ℓ_2 distances of AEs. In order to promote fairness in the community, we naturally but necessarily advocate for a fairer evaluation of the attack performance by maintaining the ℓ_∞ and ℓ_2 distances the same in both cases.

2 Related Work

2.1 Adversarial Attack

In general, the adversarial attack can be described as

$$\begin{aligned} & \text{find } \Delta \mathbf{x}, \\ & \text{s.t. } f(\mathbf{x}) \neq f(\mathbf{x} + \Delta \mathbf{x}), \\ & \|\Delta \mathbf{x}\|_p \leq \varepsilon, \end{aligned} \tag{1}$$

where $f : \Omega \subset \mathbb{R}^n \mapsto y$ refers to the victim model which predicts differently on the original samples $\mathbf{x}_{\text{org}} = \mathbf{x}$ and AEs $\mathbf{x}_{\text{adv}} = \mathbf{x} + \Delta \mathbf{x}$. Here, $\Delta \mathbf{x}$ is the perturbation, which is expected to be imperceptible in human eyes. $\Delta \mathbf{x}$ is commonly restricted by $\|\cdot\|_p$, such as ℓ_1 -, ℓ_2 -, or ℓ_∞ -norm, where ℓ_∞ is the most popular choice.

In training DNNs, a loss function $L(\cdot, \cdot)$ is minimized by updating their parameters $\mathbf{w} \in \mathbb{R}^d$ via gradient descents as

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \Gamma(\nabla_{\mathbf{w}_k} L(f(\mathbf{x}|\mathbf{w}_k), y)), \quad (2)$$

where $\{\mathbf{x}, y\}$ is the training data, and $\Gamma(\cdot)$ represents an optimization algorithm.

Adversarially, attackers maximize the loss function by taking the input $\mathbf{x}$ as the optimization variables, e.g., as follows,

$$\mathbf{x}_{k+1} = \text{clip}_{\mathbf{x}_{\text{org}}}^{\varepsilon} (\mathbf{x}_k + \Gamma(\nabla_{\mathbf{x}_k} L(f(\mathbf{x}_k|\mathbf{w}), y))), \quad (3)$$

where $\text{clip}(\cdot)$ denotes that the values of $\mathbf{x}$ (e.g., pixels in an image) are clipped to suffice the bounding restriction ε in (1). Such gradient-based attacks (Goodfellow et al., 2015; Madry et al., 2018) are very fast and thereby commonly used.

2.2 From White-Box to Black-Box

When the complete knowledge (mostly the gradients) of the victim DNNs is known, attackers are capable of achieving high success rates as described above, among which the PGD (Madry et al., 2018) is recognized as one of the strongest white-box attacks. PGD iteratively updates

$$\mathbf{x}_{k+1} = \text{clip}_{\mathbf{x}_{\text{org}}}^{\varepsilon} (\mathbf{x}_k + \alpha \cdot \text{sign}(\nabla_{\mathbf{x}_k} L(f(\mathbf{x}_k|\mathbf{w}), y))),$$

where α is the attack step size, corresponding to the learning rate in typical gradient descent. PGD simply adopts $\Gamma(\cdot) = \alpha \cdot \text{sign}(\cdot)$ but achieves a good success rate.

However, in real-world scenarios, it is commonly impossible to attain the gradient information from the victim model, and thus black-box attacks become essential. In the

existing works, black-box attacks either conduct in query-based manners (Cheng et al., 2019; Yang et al., 2020; Chen et al., 2021b) or with transfer-based approaches (Papernot et al., 2016, 2017). For the transfer-based methods we concern about in this paper, white-box attacks are carried out on a well-designed surrogate model, and the resulting AEs are expected to cheat other unknown models. To this end, various techniques have been proposed and validated effective in enhancing the attack transferability.

2.3 Transfer-Based Attacks

Optimization Algorithms It has been shown that adopting proper optimization algorithms in modifying AEs results in significantly better attack transferability. The Momentum Iterative (MI) (Dong et al., 2018) method first uses the momentum gradients, and later the Nesterov Iterative (NI) (Lin et al., 2020) chooses Nesterov accelerated gradients. Adam has also been involved in attacks recently (Yin et al., 2020).

Data Augmentation In (Xie et al., 2019b), the Diverse Input (DI) is proposed to produce AEs using average gradients from randomly transformed input samples. The Translation Invariant (TI, (Dong et al., 2019)) rather aims to generate AEs with respect to translated copies of the original samples. The Scale Invariant (SI, (Lin et al., 2020)) updates AEs with average gradients from different scale copies of the input.

Special Losses (Chen et al., 2021a) and (Wu et al., 2020) parallelly propose to Attack on Attention (AoA), which aims to suppress the attention heat map along with the cross-entropy. The heat map is a common feature across models, so those attacks on the attention boost the transferability.

Proper Back-Propagation (Wu et al., 2019) exploits the similarity between architectures with residual blocks and attacks by weighted gradients when back-propagating through the residual module. (Guo et al., 2020) proposes to remove the activation function when back-propagating the attack gradients.

3 Evaluation Method

Since our investigation is mostly empirical, we first introduce the overall evaluation method in this section, including the studied attacks and the experiment setup.

3.1 Studied Transfer-Based Attacks

We investigate 7 transfer-based attacks, which have been verified effective concerning transferability. More importantly, they naturally lead to very different ℓ_2 distances by using different optimization algorithms (on AEs) so that we could see the impacts of ℓ_2 norms clearly. These attacks could be uniformly described as (3), where $\Gamma(\cdot)$ takes the gradients $\mathbf{g}(\mathbf{x}_k)$ as the input, and outputs the optimization direction $\mathbf{g}_{k+1}$ to update as in (4). The difference among these attacks are on the optimization as specified later.

In attack, we aim to maximize the loss function $L(\cdot, \cdot)$ by optimizing the variable $\mathbf{x}$ with gradients $\mathbf{g}(\mathbf{x}) = \nabla_{\mathbf{x}} L(f(\mathbf{x}|\mathbf{w}), y)$. To guarantee the imperceptibility (e.g., the pixel-wise RGB restrictions for images), $\mathbf{x}$ is ℓ_∞ -bounded by threshold ε around the original sample $\mathbf{x}_{\text{org}}$ as

$$\mathbf{x}_{k+1} = \text{clip}_{\mathbf{x}_{\text{org}}}^{\varepsilon} (\mathbf{x}_k + \alpha \cdot \mathbf{g}_{k+1}). \quad (4)$$

In the past, different attacks are restricted to the same ℓ_∞ bound, for a fair comparison.

Gradient Descent (GD) (Madry et al., 2018)

$$\mathbf{g}_{k+1} = \mathbf{g}(\mathbf{x}_k).$$

Gradient descent algorithm performs the updates through the computed gradient direction. The corresponding attack is one-step FGSM (Goodfellow et al., 2015) and multi-step PGD (Madry et al., 2018).

Momentum Gradient Descent (M-GD) (Dong et al., 2018)

$$\mathbf{g}_{k+1} = \mu \cdot \mathbf{g}_k + \mathbf{g}(\mathbf{x}_k).$$

M-GD accumulates the gradients in a moving-average way, so it implicitly optimizes with more one-stage information as in GD. As a result, it gains a faster convergence and reduces oscillation.

Nesterov Accelerated Gradient Descent (NA-GD) (Lin et al., 2020)

$$\mathbf{g}_{k+1} = \mu \cdot \mathbf{g}_k + \mathbf{g}(\mathbf{x}_k + \alpha \cdot \mu \cdot \mathbf{g}_k).$$

NA-GD makes a big jump in the previously accumulated gradient and does a correction afterward. This predictive strategy prevents it from going too fast and leads to increased responsiveness.

Adam (Kingma and Ba, 2015)

$$\mathbf{m}_{k+1} = \beta_1 \cdot \mathbf{m}_k + (1 - \beta_1) \cdot \mathbf{g}(\mathbf{x}_k),$$

$$\boldsymbol{\nu}_{k+1} = \beta_2 \cdot \boldsymbol{\nu}_k + (1 - \beta_2) \cdot \mathbf{g}^2(\mathbf{x}_k),$$

$$\mathbf{g}_{k+1} = \frac{\mathbf{m}_{k+1}}{\sqrt{\boldsymbol{\nu}_{k+1} + \delta}},$$

where the squared root $\sqrt{(\cdot)}$ is performed element-wisely. Besides storing a moving average of past squared gradients, Adam also keeps that of past gradients as in M-GD.

The crucial idea is to go slower in dimensions that have gone far.

L-Adam (Loshchilov and Hutter, 2017)

$$\mathbf{m}_{k+1} = \beta_1 \cdot \mathbf{m}_k + (1 - \beta_1) \cdot (\mathbf{g}(\mathbf{x}_k) - \lambda \cdot \mathbf{x}_k),$$

$$\boldsymbol{\nu}_{k+1} = \beta_2 \cdot \boldsymbol{\nu}_k + (1 - \beta_2) \cdot (\mathbf{g}(\mathbf{x}_k) - \lambda \cdot \mathbf{x}_k)^2,$$

$$\mathbf{g}_{k+1} = \frac{\mathbf{m}_{k+1}}{\sqrt{\boldsymbol{\nu}_{k+1}} + \delta}.$$

L-Adam adopts λ to regularize the gradients. The ℓ_2 regularization is very frequently referred to as the weight decay.

AdaBelief (Zhuang et al., 2020)

$$\mathbf{m}_{k+1} = \beta_1 \cdot \mathbf{m}_k + (1 - \beta_1) \cdot \mathbf{g}(\mathbf{x}_k),$$

$$\boldsymbol{\nu}_{k+1} = \beta_2 \cdot \boldsymbol{\nu}_k + (1 - \beta_2) \cdot (\mathbf{g}(\mathbf{x}_k) - \mathbf{m}_{k+1})^2,$$

$$\mathbf{g}_{k+1} = \frac{\mathbf{m}_{k+1}}{\sqrt{\boldsymbol{\nu}_{k+1}} + \delta}.$$

Compared to Adam, AdaBelief takes a large step when $\mathbf{g}_k$ is close to $\mathbf{m}_k$, and a small step when $\mathbf{g}_k$ greatly deviates from $\mathbf{m}_k$. AdaBelief has better convergence on many tasks with no extra costs.

M-SVAG (Balles and Hennig, 2018)

$$\begin{aligned}\mathbf{m}_{k+1} &= \beta \cdot \mathbf{m}_k + (1 - \beta) \cdot \mathbf{g}(\mathbf{x}_k), \\ \boldsymbol{\nu}_{k+1} &= \beta \cdot \boldsymbol{\nu}_k + (1 - \beta) \cdot \mathbf{g}^2(\mathbf{x}_k), \\ \rho_{k+1} &= \frac{(1 - \beta)(1 + \beta^{k+2})}{(1 + \beta)(1 - \beta^{k+2})}, \\ \mathbf{g}_{k+1} &= \frac{(1 - \rho_{k+1}) \cdot \mathbf{m}_{k+1}^2}{(1 - 2\rho_{k+1}) \cdot \mathbf{m}_{k+1}^2 + \rho_{k+1} \cdot \boldsymbol{\nu}_{k+1}} \cdot \mathbf{m}_{k+1}.\end{aligned}$$

Momentum Stochastic Variance-Adapted Gradient (M-SVAG), compared to M-GD, applies variance adaptation to update the direction. The formula above has been transformed without change.

3.2 Experimental Setup

In all studies, ImageNet (Deng et al., 2009) validation set with 50K images are used as original samples, following (Xie et al., 2019b; Lin et al., 2020; Chen et al., 2021a). Keras pre-processing function, central cropping, and resizing (to 224 or 299) are adopted to pre-process the inputs. In all experiments, 1K images are randomly (with a fixed seed) selected and the images falsely predicted by the surrogate are skipped as in (Su et al., 2018; Chen et al., 2021a). Experiments are implemented in TensorFlow (Abadi et al., 2016), Keras (Chollet et al., 2015) with 4 NVIDIA GeForce RTX 2080Ti GPUs.

For attack and test, several well-trained models in Keras Applications are used. VGG16 (Simonyan and Zisserman, 2015), ResNet50 (RN50) (He et al., 2016), InceptionV3 (IncV3) (Szegedy et al., 2016), and DenseNet121 (DN121) (Huang et al., 2017) are attacked as surrogate models. VGG19 (Simonyan and Zisserman, 2015), ResNet152 (RN152) (He et al., 2016), DenseNet201 (DN201) (Huang et al., 2017), InceptionResNetV2 (IncRN) (Szegedy et al., 2017), Xception (Xcpt) (Chollet, 2017), and

NASNetLarge (NASL) (Zoph et al., 2018) are chosen as black-box victims. To enrich the comparison, we also test on some adversarially-trained (not by our attack) black box models, including InceptionV3adv (A1, (Kurakin et al., 2017)), InceptionResNetV2adv (A2, (Kurakin et al., 2017)), and ResNeXt101den (A3, (Xie et al., 2019a)). The extreme diversity of models eliminates the biases from model architectures to a large extent, contributing to the solid conclusion concerning attacks and ℓ_2 distances.

The Attack Success Rate (ASR) is taken as the performance metric, which reflects the transferability and is measured by the average error rate of 9 victim models. Since the ℓ_2 distance is dependent on the size on images, we use a normalized metric to represent its value, which is the Root Mean Square Error (RMSE). RMSE calculates the ℓ_2 distance that AEs go away as $dist(\mathbf{x}_{adv}, \mathbf{x}_{org}) = \sqrt{\|\mathbf{x}_{adv} - \mathbf{x}_{org}\|_2^2 / N}$, where $N = H \times W \times C$ is the dimension of the sample.

To fairly evaluate the transferability of attacks, the optimization problem of the adversarial attack on the surrogate model is solved to a relatively equal degree with updates presented in Section 3.1. That is to say, the loss of the surrogate on AEs is controlled to a fixed value (such as 0.03 in our experiments if not otherwise stated), of which the methodological details are described in Appendix A with all other hyperparameters. All perturbations are ℓ_∞ -bounded by $\varepsilon = 16$ following (Xie et al., 2019b; Dong et al., 2019; Lin et al., 2020).

4 ℓ_2 Distance Affects ℓ_∞ Attacks

In this section, we reveal the current insufficient measurement of the attack strength by showing the influence of ℓ_2 norm on the transferability via experiments on 7 transfer-based attacks, 4 surrogates, and 9 victims. Under the same ℓ_∞ bound, 1) attacks that optimize AEs to larger ℓ_2 distances enjoy better transferability; 2) enlarging ℓ_2 distances of AEs leads to a steady increase of transferability; 3) fixing the ℓ_2 distances of AEs produces amazingly consistent transferability for all attacks. Observations above solidly validate the deficiency of current attack strength measurements by solely the ℓ_∞ norm.

4.1 Transferability Correlates with RMSE

We first empirically study the transferability under the above basic setting. Results are plotted in Fig. 1, where two indices are worth mentioning and comparing, i.e., the ASR (transferability) and the RMSE of AEs generated by the tested 7 attacks.

Looking at the lines (for transferability) and bars (for RMSE) in the same color simultaneously, one could observe that the transferability has a strong correlation with the RMSE. For the transferability, Adam family outperforms the others, and then M-SVAG follows. More specifically, the resulting transferability is ranked: Adam family > M-SVAG > GD > NA-GD $\approx$ M-GD. For the RMSE, the Adam family also stands out, and the rank is similar: Adam family > GD > M-SVAG > NA-GD $\approx$ M-GD. Among the surveyed attacks, except for a slight difference concerning GD, this phenomenon holds true for the rest of the attacks. More notably, this overall trend remains stable for all 4 surrogates.

The results show that when the attack is equally conducted on surrogates, the ASR

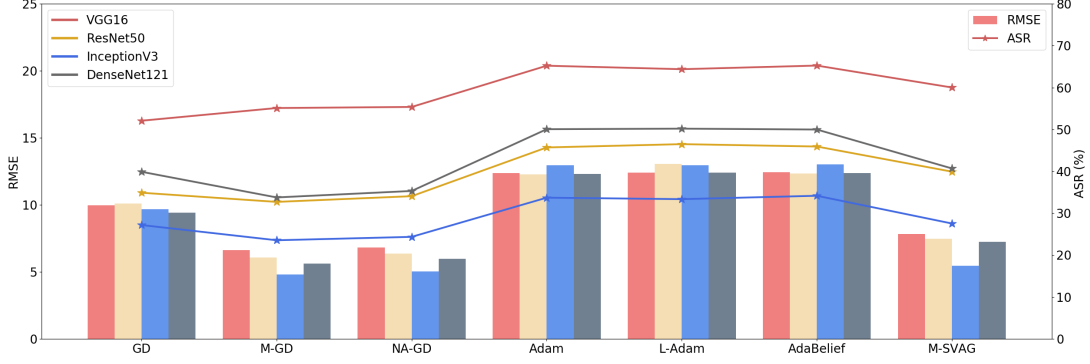


Figure 1: Transferability (lines) and RMSE (bars) of the AEs crafted by 7 attacks, where the cross-entropy loss of the surrogate model are maintained to 0.03 for fair comparisons. For lines and bars, colors indicate surrogates.

(transferability) of a certain attack has a strong correlation with the corresponding RMSE between AEs and original samples, indicating that RMSE may be an important factor for the transferability. To further validate this hypothesis, we consider directly intervening in the RMSE and to see the effect on the transferability.

4.2 Transferability with Larger RMSE

If the hypothesis is correct, then by straightforward increasing RMSE while keeping other settings the same, the transferability would be boosted. RMSE can be enlarged in several ways. Here, we offer one possible strategy, which is to add an RMSE item to the original attack loss as the following,

$$L'(f(\mathbf{x}|\mathbf{w}), y) = L(f(\mathbf{x}|\mathbf{w}), y) + \gamma \cdot \text{dist}(\mathbf{x}, \mathbf{x}_{\text{org}}), \quad (5)$$

where $L(f(\mathbf{x}|\mathbf{w}), y)$ is the standard cross-entropy loss, $\text{dist}(\mathbf{x}, \mathbf{x}_{\text{org}})$ is the added RMSE between the potential AEs and their corresponding original samples, and the parameter $\gamma > 0$ controls the trade-off between the two terms.

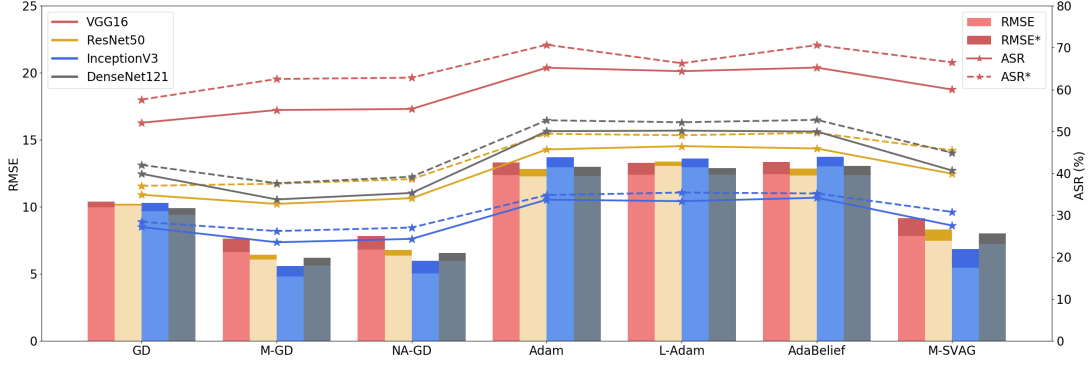


Figure 2: Transferability (lines) and RMSE (bars) of the AEs crafted by 7 attacks with a constant cross-entropy loss of 0.03 in the surrogates and $\gamma = 0.004$ in the modified loss (5). Full lines and light bars are reference results from Fig. 1. Dotted lines (ASR*) and the shadowed dark bars (RMSE*, on top of the bars) are results with the additional RMSE item in loss, where the AEs reach a larger RMSE when the attack stops at the same cross-entropy loss.

With the modified loss function (5), the influence of RMSE item $dist(\mathbf{x}, \mathbf{x}_{org})$ can now be evaluated in a quantitative way, maintaining the cross-entropy item $L(f(\mathbf{x}|\mathbf{w}), y)$ to a fixed value. Under this setting, we extend the experiment of Fig. 1 by performing the attacks with the modified loss (5) until the cross-entropy item $L(f(\mathbf{x}|\mathbf{w}), y)$ reaches the same level as that in Fig. 1, such that the RMSE is the only factor that varies. We can see from Fig. 2 that a definite boost in transferability (from the full lines copied from Fig. 1 to the dotted lines with (5)) is obtained under all circumstances, validating our hypothesis.

4.3 Transferability with Fixed RMSE

Besides adding an item to induce a large RMSE, another possible validation is to directly control the RMSE to the same level, which of course would produce a similar

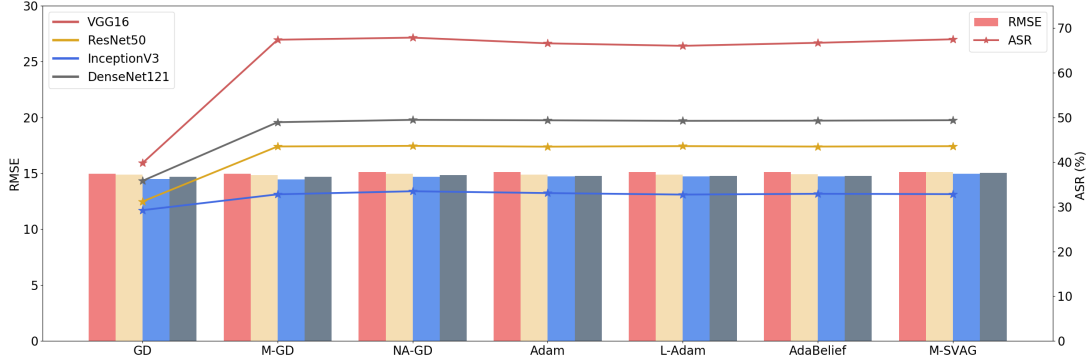


Figure 3: Transferability (lines) and RMSE (bars) of the AEs crafted by 7 attacks with a constant RMSE, i.e., $dist(\mathbf{x}, \mathbf{x}_{org}) = 15$.

ASR if our hypothesis is correct. Because if attackers influence the transferability via RMSE, then fixing RMSE would eliminate this influence chain, leading to a fixed ASR.

Therefore, we carefully tune the hyper-parameters as specified in Appendix A to maintain AEs to have a similar RMSE $dist(\mathbf{x}, \mathbf{x}_{org})$ by stopping attack when AEs approach a pre-set RMSE value 15 and report the results in Fig. 3. Amazingly, the transferability goes into a similar value for all surrogates and even almost all attacks. The 4 flat lines indicate that the ASR towards 9 victims is found alike however the specific optimization proceeds. Because the RMSE, is controlled unchangeable, different optimization algorithms could hardly impact the transferability, which is revealed and demonstrated by nearly all performed studies.

This encouraging discovery also positively supports our hypothesis that attackers could impact the transferability via RMSE because when the RMSE is not allowed to change, attacks do not work in boosting the transferability just like the powerless curves failing to oscillate in Fig. 3. All pre-average results are presented in Appendix B.

5 Discussions on Fair Evaluations by ℓ_2 Norms

Previously, we have empirically validated that ℓ_2 distances (measured by RMSE) greatly influence the attack performance, and thus is an important metric for the attack strength. In this section, we demonstrate that the seemingly heuristic and intuitive conclusion is beneficial and inspirational to the community. We provide more examples with explanations to highlight the current lack of fair evaluations, use a counter-example to show the importance of ℓ_2 measurements, and thus advocate for proper evaluations of the attack strength by both the widely used ℓ_∞ bound and the ℓ_2 distances (RMSE) to avoid trivial improvements in the attack transferability.

5.1 “Boosting” Transferability by Larger Perturbations

Although it is obvious that under the same ℓ_∞ bound, the larger-perturbed AEs according to another metric (the ℓ_2 norm), have greater transferability, we provide explanations here on how and why this phenomenon is prevalent (non-preferable to enhance transferability in this way due to the perceptibility of AEs).

The deep reason for this is that the attack gradients is mostly sparse, i.e., extremely large in few pixels and small in most pixels, which could be observed in the ℓ_2 attacks as in Fig. 4 because the actual gradient values are preserved there. Due to the sparsity of attack gradients, the sign function in common ℓ_∞ attacks frequently leads to zero attack updates, seeing a numerical result here. PGD generally produces AEs with RMSE around 7 under the ℓ_∞ bound $\varepsilon = 16$, which is far from the maximum RMSE distance of 16 if all pixels are fully perturbed without zero updates.

Because there are so many insufficiently perturbed pixels in PGD, methods may



Figure 4: AEs of ℓ_2 attacks. The sparsity of attack gradients is observable here, i.e., the strength of perturbations (proportional to the actual attack gradients) varies a lot in different areas. The AEs are crafted by attacking VGG19 (Simonyan and Zisserman, 2015) by SI-AoA (Chen et al., 2021a; Lin et al., 2020).

well implicitly enhance the transferability by increasing the attack strength, which is however not preferred. For example, MI-PGD (Dong et al., 2018) and NI-PGD (Lin et al., 2020) mostly produce AEs with RMSE around 12, which is significantly larger than that in PGD, making the AEs more perceivable. Another example is the recent state-of-the-art query-based attack, the Square Attack (Andriushchenko et al., 2020), which generates the maximum-perturbed (under the ℓ_∞ bound) AEs for each query, so its great performance may be partially due to this exhausted perturbing strategy. We visualize AEs from small perturbed attacks and large perturbed ones in Fig. 5, where one could observe that the attack strength and the perceptibility of AEs differ a lot even they are bounded under the same ℓ_∞ norm.

Additionally, it has been pointed out (Su et al., 2018) that FGSM (Goodfellow et al., 2015) transfers better than PGD (Madry et al., 2018), but this is a bit contradictory with the common fact that PGD is stronger in white-box settings and produces more diverse patterns, indicating good transferability (Chen et al., 2021a). Now based on

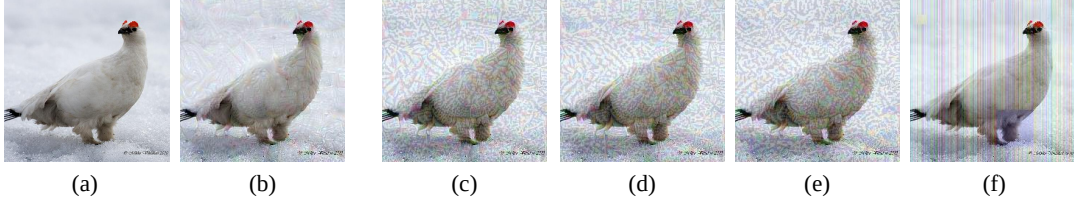


Figure 5: AEs from small-perturbed (b) and large-perturbed attacks (c), (d), (e), (f). (a) is the original image; (b) is the PGD AE; (c) is the MI-PGD AE; (d) is the NI-PGD AE; (e) is the FGSM AE; (f) is the Square Attack AE. The AEs are crafted by attacking VGG16 (Simonyan and Zisserman, 2015) within ℓ_∞ bound 16.

our conclusion, this phenomenon could be well explained. FGSM perturbs pixels only once, so it extremely exploits the feasible values within the ℓ_∞ bound to push every pixel to the limit, leading to the maximum RMSE because the attack gradients in the first iteration have not been very sparse. The brute strategy of FGSM works better than PGD, implying that the impact of ℓ_2 distances is even more significant than that of precise updates of AEs (in PGD), when it comes to the attack transferability.

5.2 Fairness Matters: A Simple Counter-Example Attack

We provide a typical but simple counter-example to illustrate the unfairness caused by the single measurement of the ℓ_∞ norm. Our counter-example attack stops the attack iteration only when RMSE starts to decrease rather than after a certain number of iterations. We conduct experiments with a common step size $\alpha = 2$ of the sign update, and report the results in Table 1, which shows that such a simple, trivial, and naive strategy should increase the transferability by 20%. That is, if we lack a proper measure of the attack strength, attackers could produce stronger and more perceivable AEs under the same ℓ_∞ without modifying the method as in Fig. 6. More results are in Appendix C.

Table 1: ASR of the Counter-Example Attack (✓) Compared to Baselines (×)

Surrogate / Baseline		Undefended Victim						Defended Victim		
		DN201	VGG19	RN152	IncRN	Xcpt	NASL	A1	A2	A3
VGG16	×	58.0	91.2	40.7	37.6	44.4	35.1	35.4	29.6	16.3
DI-TI	✓	69.9	96.6	48.9	47.6	53.1	50.8	43.9	38.1	17.0
VGG16	×	67.2	94.8	52.1	47.2	49.3	40.0	47.7	40.9	20.2
DI-TI-MI	✓	77.7	98.1	56.7	54.7	60.6	54.1	53.1	45.2	17.8
RN50	×	61.4	62.3	42.5	35.2	34.8	30.4	34.3	29.1	17.0
DI-TI	✓	72.6	69.9	50.6	43.1	41.0	38.9	40.5	36.9	17.6
RN50	×	72.4	74.4	55.6	47.1	48.2	40.0	47.4	40.3	20.7
DI-TI-MI	✓	80.1	79.7	60.7	53.4	51.5	45.6	51.8	46.3	20.7
IncV3	×	40.4	49.2	38.4	35.8	32.2	22.7	24.3	22.1	22.2
DI-TI	✓	49.5	59.2	45.0	45.3	38.7	29.3	32.4	27.9	23.1
IncV3	×	51.4	62.2	49.3	46.4	43.4	32.7	41.5	35.0	24.2
DI-TI-MI	✓	59.2	72.0	54.7	52.7	48.9	38.1	44.4	38.9	24.3
DN121	×	75.3	61.9	39.4	36.1	38.6	32.9	29.9	27.6	17.3
DI-TI	✓	85.3	71.8	46.4	45.1	47.1	43.7	39.8	35.8	17.4
DN121	×	80.7	74.9	51.6	46.7	48.7	42.5	46.8	39.0	21.2
DI-TI-MI	✓	92.0	80.7	58.1	56.8	57.8	54.7	54.3	47.2	19.2

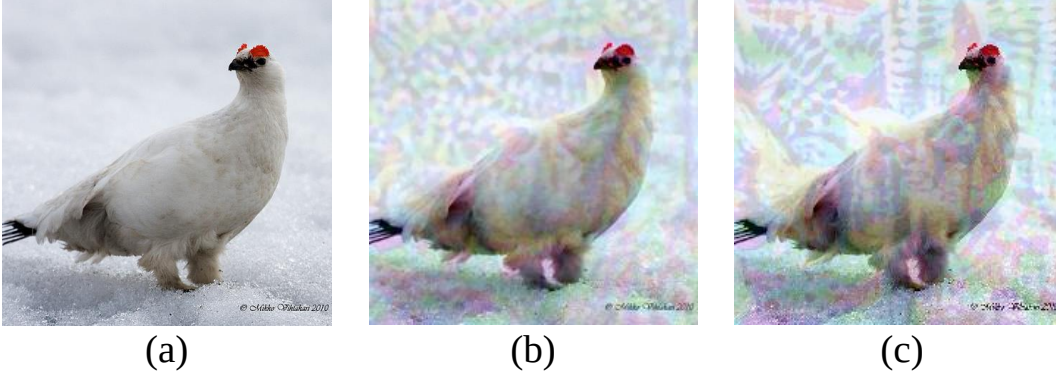


Figure 6: AEs of our counter-example method and the baseline. (a) is the original image; (b) is the DI-TI-MI-PGD AE; (c) is our counter-example AE using DI-TI-MI-PGD. The AEs are crafted by attacking VGG16 within ℓ_∞ bound 16.

5.3 Measuring the Attack Strength Properly

In most of the current attack experiments, e.g., (Dong et al., 2018; Xie et al., 2019b; Dong et al., 2019; Lin et al., 2020; Mao et al., 2021), the attack strength is measured by the ℓ_∞ norm only. However, as verified in the above experiments, one could change the ℓ_2 distance to get different attack performance within the same ℓ_∞ bound. In this concern, we advocate measuring the strength of all attacks by both the ℓ_∞ and ℓ_2 distance. In other words, to fairly evaluate the attack transferability, one should maintain the same ℓ_2 distance under a fixed ℓ_∞ bound in ℓ_∞ attack as in (Chen et al., 2021a), and also report the RMSE as in (Liu et al., 2017; Wu et al., 2019).

Conducting a fair test on attack transferability is very important, which could prevent the researchers from explicitly or implicitly resorting to tricky ways to enhance the transferability by enlarging perturbations. With our conclusion and the advocated unprejudiced evaluations, the community should focus on the real vulnerability of DNNs, which is the intentions or meanings of adversarial attacks.

6 Conclusion

We investigate the measurement of the attack strength when evaluating the transferability, and find that existing practice to bound only the ℓ_∞ norm is insufficient. To reveal this, we show that the ℓ_2 distance of AEs, a widely neglected metric in ℓ_∞ attacks, also exerts a great impact on the resulting transferability from comprehensive experiments. Accordingly, attackers are supposed to at least consider the ℓ_2 distances as well in all attacks, otherwise, it is possible to bypass the measurement in a tricky and trivial way to seemingly enhance the transferability by actually increasing the attack strength, which makes the AEs more perceivable. Our work is only the first step for the fair evaluation since ℓ_2 norm is just one metric besides the restricted ℓ_∞ norm, and other metrics may be also important, e.g., the ℓ_0 norm, the ℓ_1 norm, SSIM, the distance in feature spaces.

Acknowledgements

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A Implement Details

Transfer-Based Attacks via Optimization We mostly adopt the hyper-parameters recommended by the proposers. For M-GD, μ is set to 0.9. For Adam, the authors suggest $\beta_1 = 0.9$ and $\beta_2 = 0.999$ as a mild choice. To encourage a fast attack, we eliminate the bias correction term herein. Since $\mathbf{g}$ is extremely smaller than $\mathbf{x}$ in attack, we choose $\lambda = 10^{-12}$ in L-Adam. M-SVAG sets $\beta_1 = \beta_2 = \beta = 0.9$. For Adam family, we select $\delta = 10^{-8}$ to avoid zero division.

Other Transfer-Based Attacks DI transforms the image for 4 times with probability $p = 1$. The transformation is to resize the image to $0.9 \times$ its size and randomly padding the outer areas with white pixels. SI divides the sample numerically by the power 2 for 4 times. TI adopts a kernel size 15 as suggested. MI uses momentum with $\mu = 1$.

Attack Step The learning rate, or attack step size denoted as α needs to be different for each sample, given the unevenly-distributed gradients across data (Xie et al., 2017). Accordingly, we need to decide on $\alpha(\mathbf{x})$ to ensure a proper degree of perturbations. Specifically, we make the perturbations after the first attack iteration $\mathbf{x}_1 - \mathbf{x}_0$ have a fixed RMSE ξ as

$$\begin{aligned} & \text{find } \alpha(\mathbf{x}), \\ & \text{s.t. } \mathbf{x}_1 = \text{clip}_{\mathbf{x}_0}^{\varepsilon}(\mathbf{x}_0 + \alpha(\mathbf{x}_0) \cdot \mathbf{g}_1), \\ & \text{dist}(\mathbf{x}_1, \mathbf{x}_0) = \xi. \end{aligned} \tag{6}$$

Without the sign operation, the optimization direction of algorithms would be precisely preserved in Section 3.

Maintain Loss and RMSE Because the loss and RMSE increase very unstably, we tune ξ for algorithms to achieve a fixed loss or RMSE of the final AEs as specified in Table 2. Attacks stop if the iteration is 10 or the AEs reach the preset loss in Section 4.1, 4.2 and RMSE in Section 4.3.

Table 2: ξ to Maintain Loss or RMSE

Algorithm	Loss	RMSE
GD	0.3	8
M-GD	0.01	2
NI-GD	0.01	2
Adam	1	4
L-Adam	1	4
AdaBelief	1	4
M-SVAG	0.01	2

B Pre-Average Results in Section 4

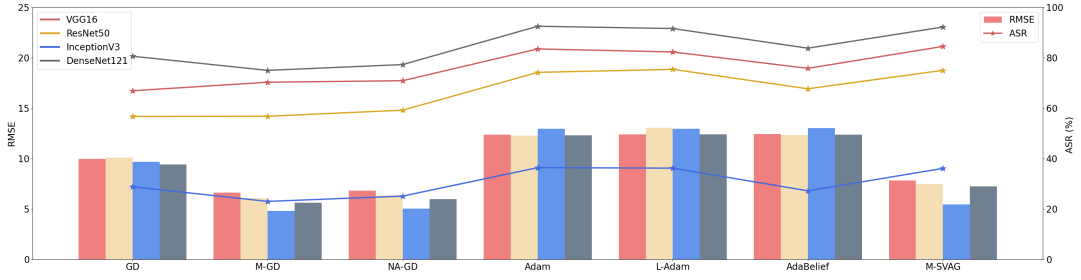


Figure 7: Transferability on DenseNet201 and RMSE of the AEs.

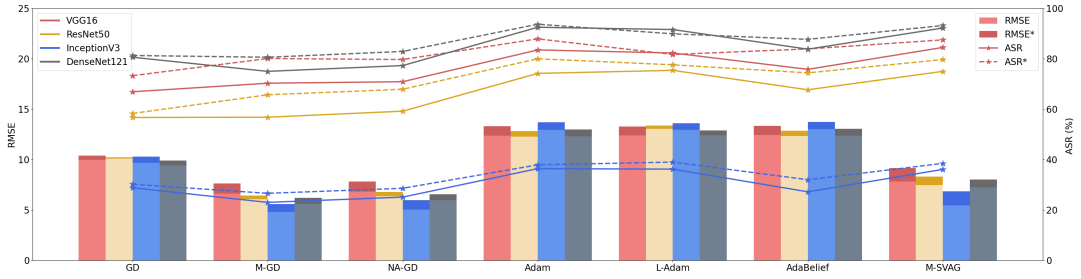


Figure 8: Transferability on DenseNet201 and RMSE of the AEs with modified loss.

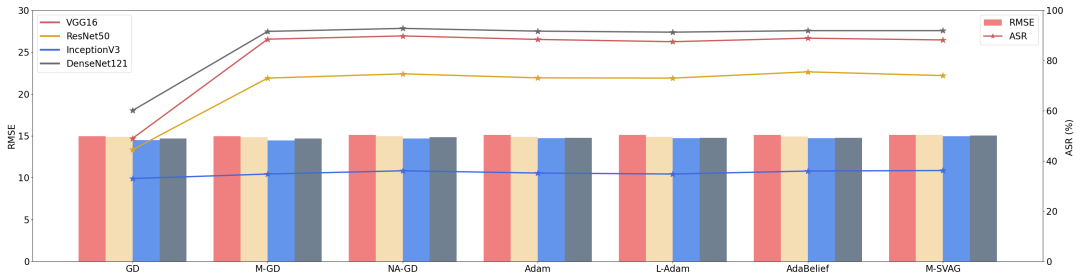


Figure 9: Transferability on DenseNet201 and RMSE of the AEs with constant RMSE.

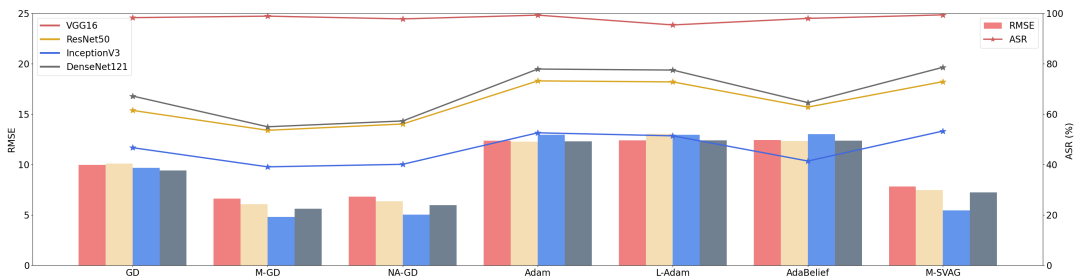


Figure 10: Transferability on VGG19 and RMSE of the AEs.

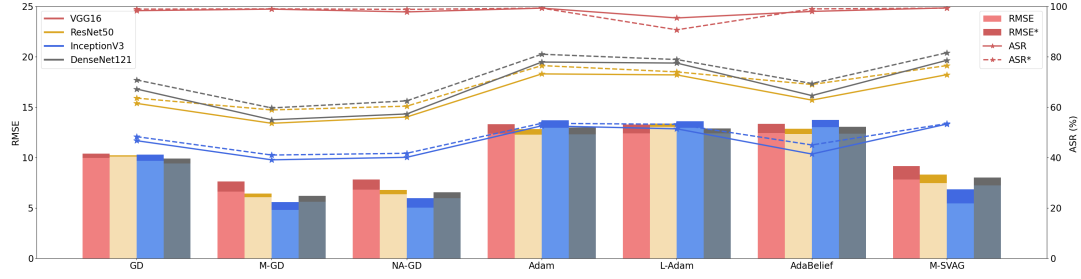


Figure 11: Transferability on VGG19 and RMSE of the AEs with modified loss.

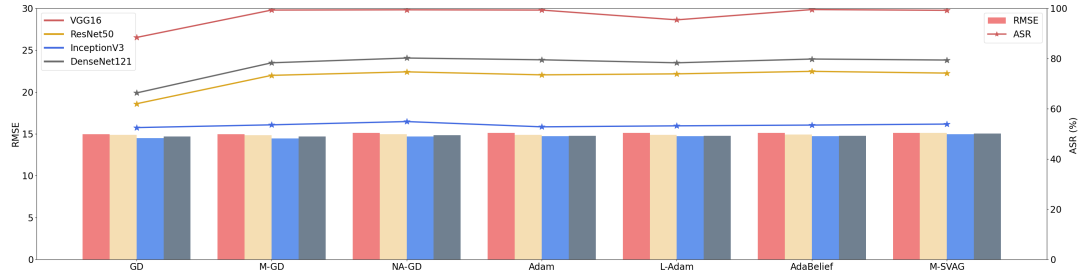


Figure 12: Transferability on VGG19 and RMSE of the AEs with constant RMSE.

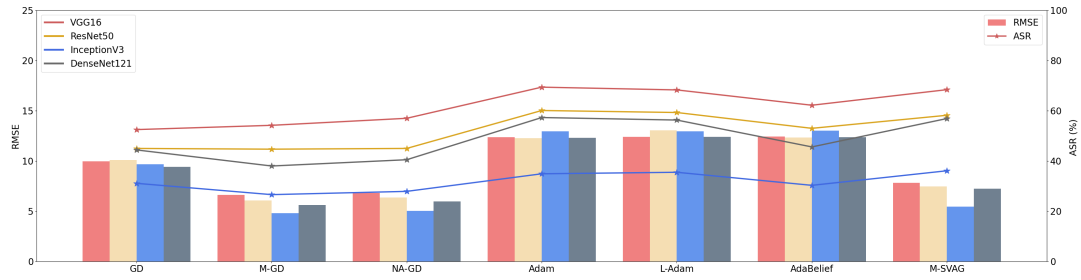


Figure 13: Transferability on ResNet152 and RMSE of the AEs.

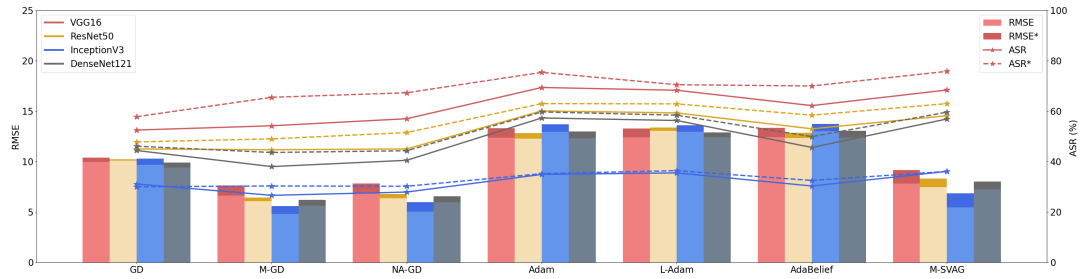


Figure 14: Transferability on ResNet152 and RMSE of the AEs with modified loss.

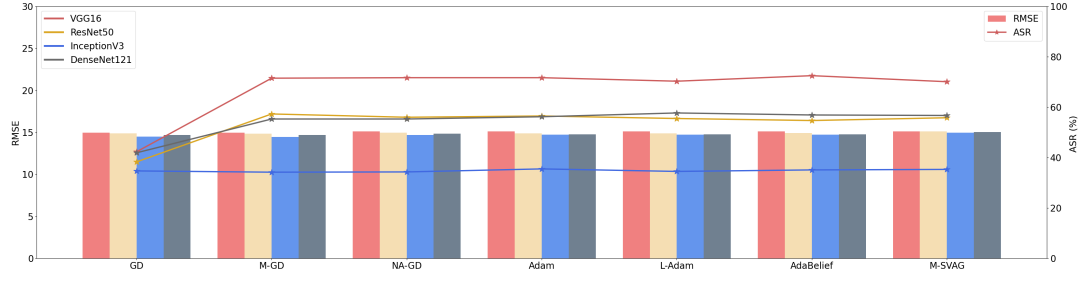


Figure 15: Transferability on ResNet152 and RMSE of the AEs with constant RMSE.

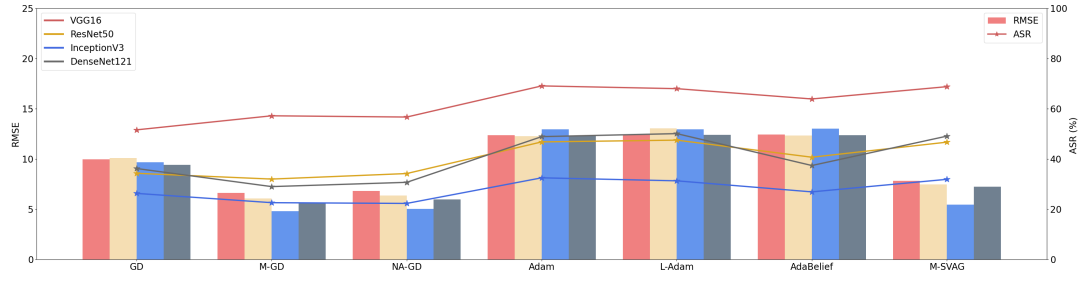


Figure 16: Transferability on InceptionResNetV2 and RMSE of the AEs.

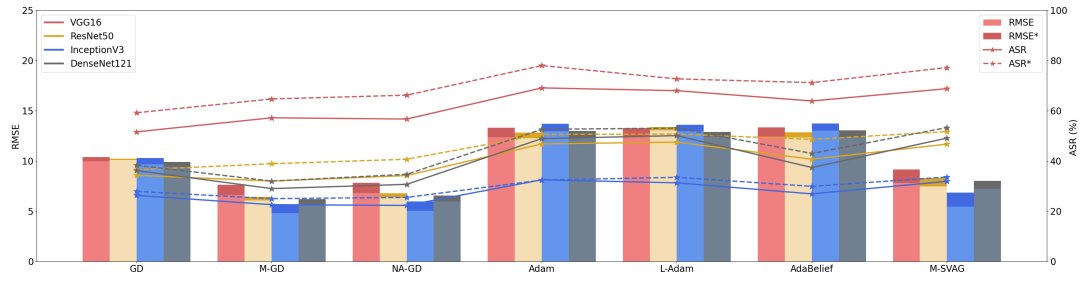


Figure 17: Transferability on InceptionResNetV2 and RMSE of the AEs with modified loss.

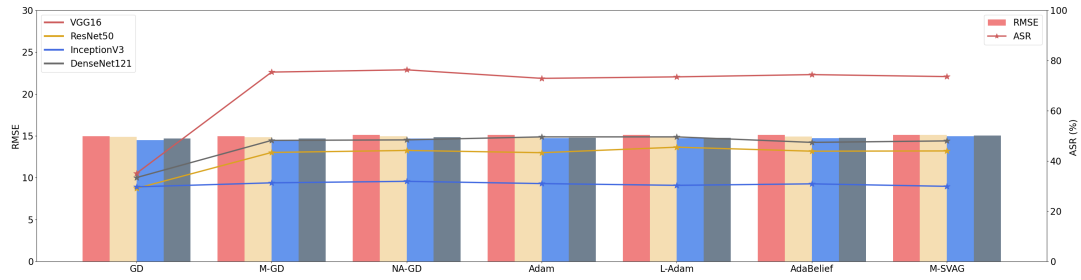


Figure 18: Transferability on InceptionResNetV2 and RMSE of the AEs with constant RMSE.

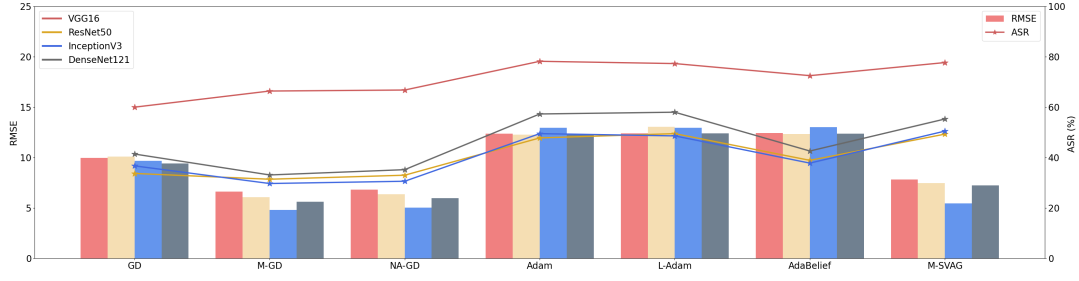


Figure 19: Transferability on Xception and RMSE of the AEs.

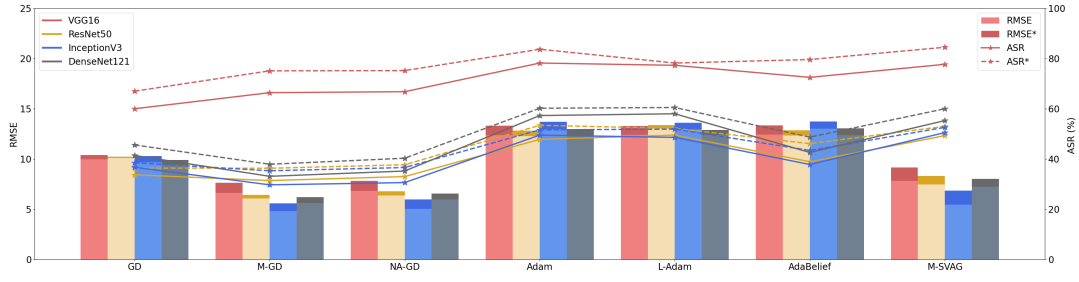


Figure 20: Transferability on Xception and RMSE of the AEs with modified loss.

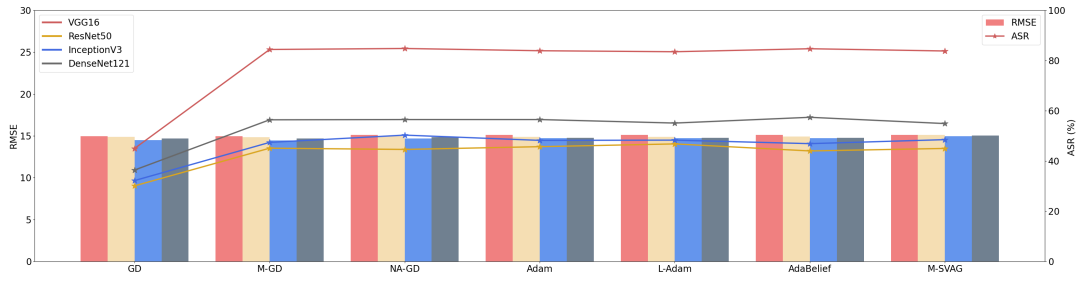


Figure 21: Transferability on Xception and RMSE of the AEs with constant RMSE.

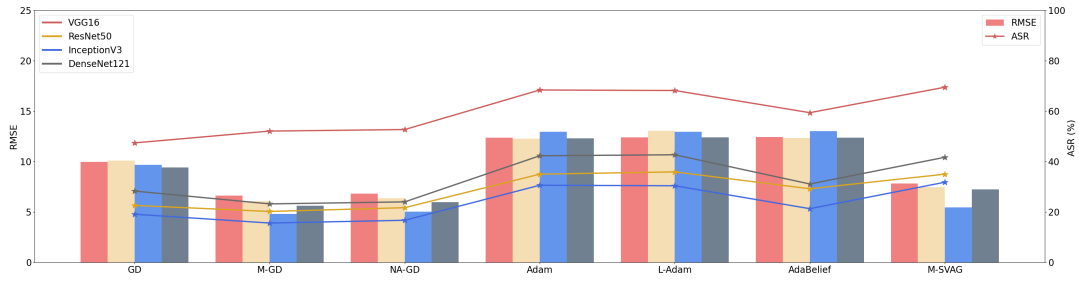


Figure 22: Transferability on NASNetLarge and RMSE of the AEs.

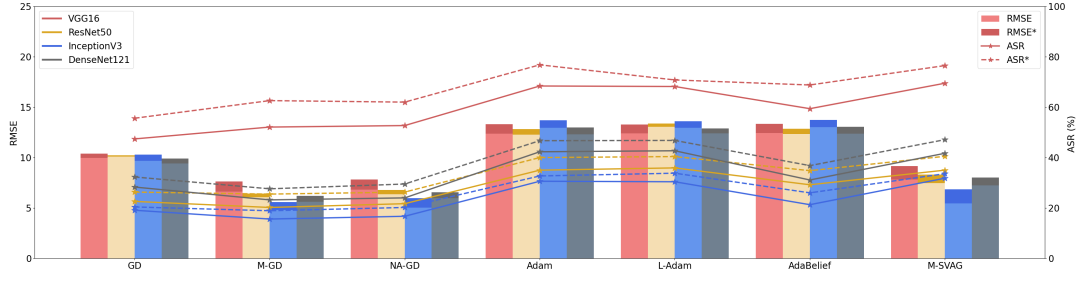


Figure 23: Transferability on NASNetLarge and RMSE of the AEs with modified loss.

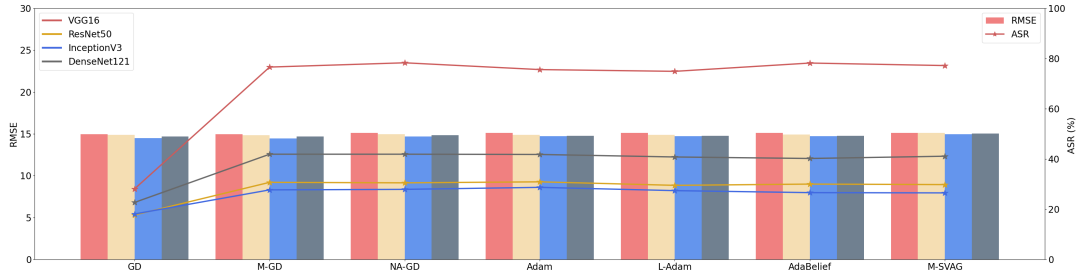


Figure 24: Transferability on NASNetLarge and RMSE of the AEs with constant RMSE.

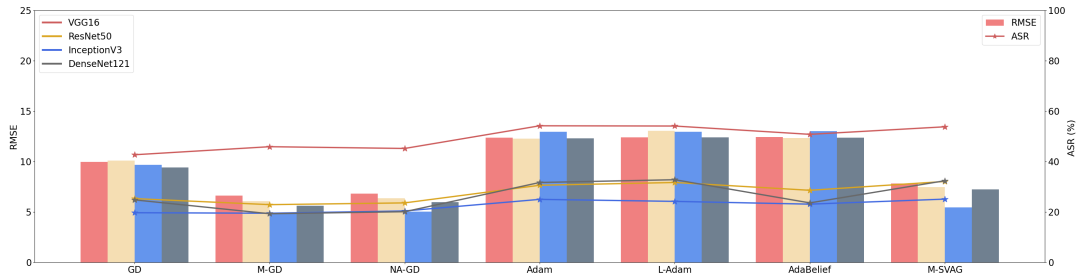


Figure 25: Transferability on InceptionV3adv and RMSE of the AEs.

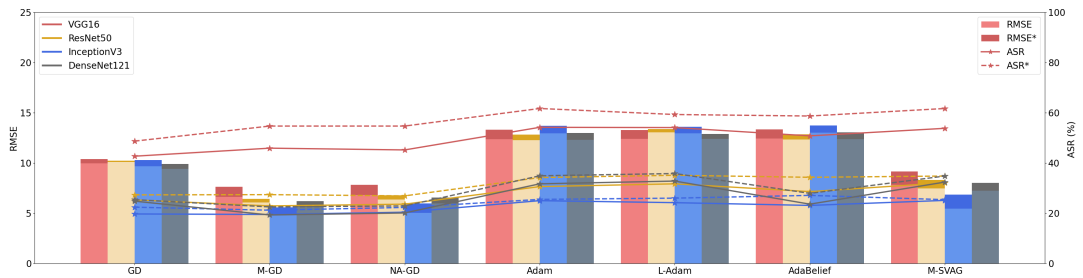


Figure 26: Transferability on InceptionV3adv and RMSE of the AEs with modified loss.

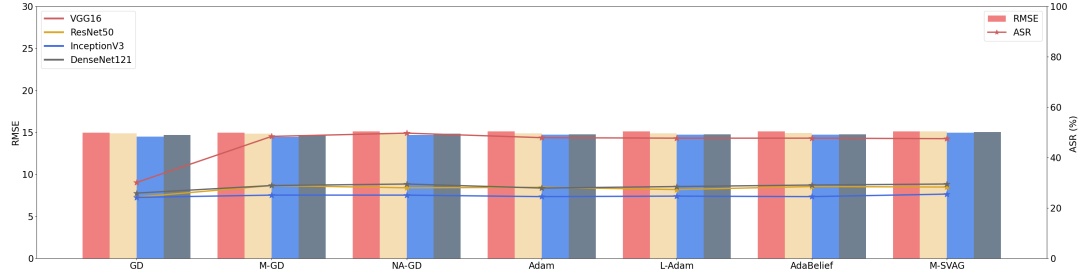


Figure 27: Transferability on InceptionV3adv and RMSE of the AEs with constant RMSE.

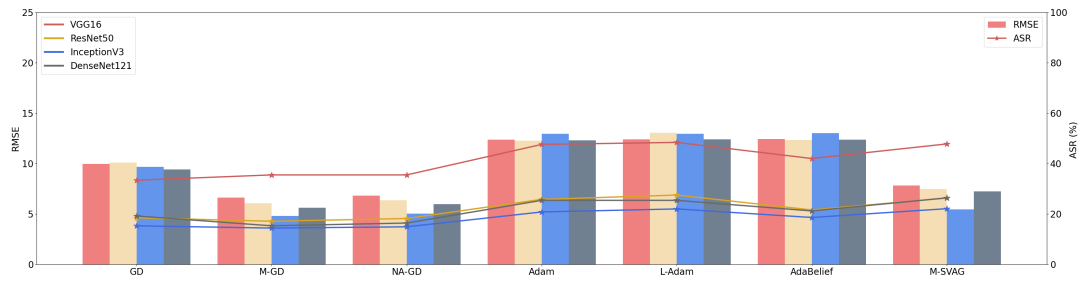


Figure 28: Transferability on InceptionResNetV2adv and RMSE of the AEs.

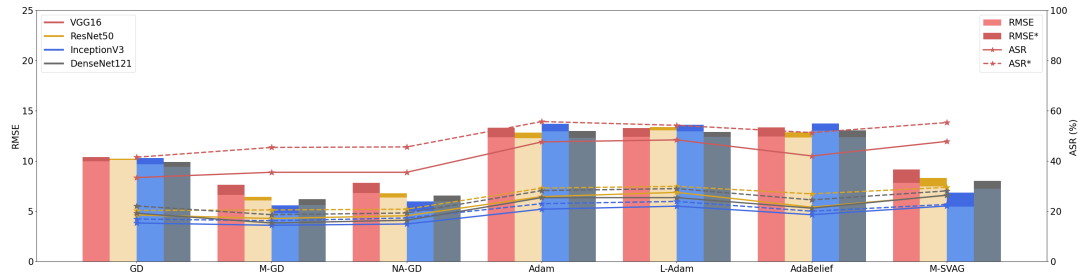


Figure 29: Transferability on InceptionResNetV2adv and RMSE of the AEs with modified loss.

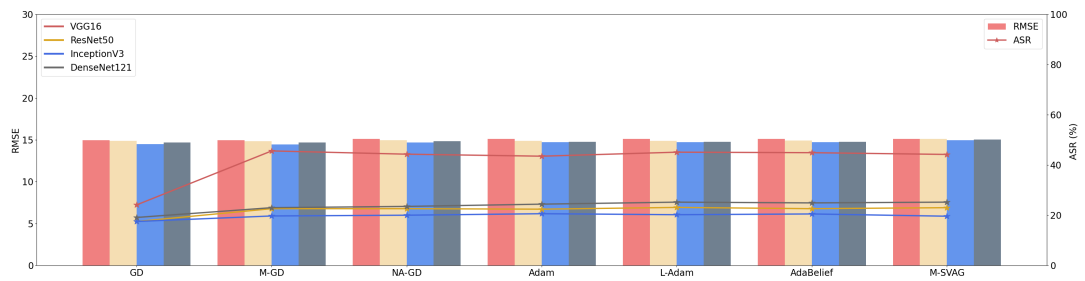


Figure 30: Transferability on InceptionResNetV2adv and RMSE of the AEs with constant RMSE.

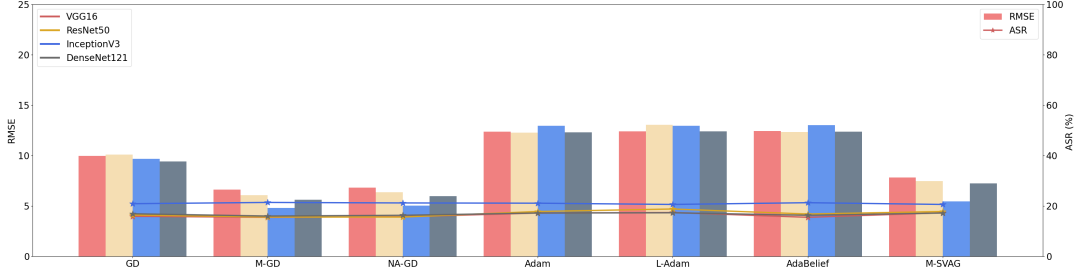


Figure 31: Transferability on ResNetXt101denoise and RMSE of the AEs.

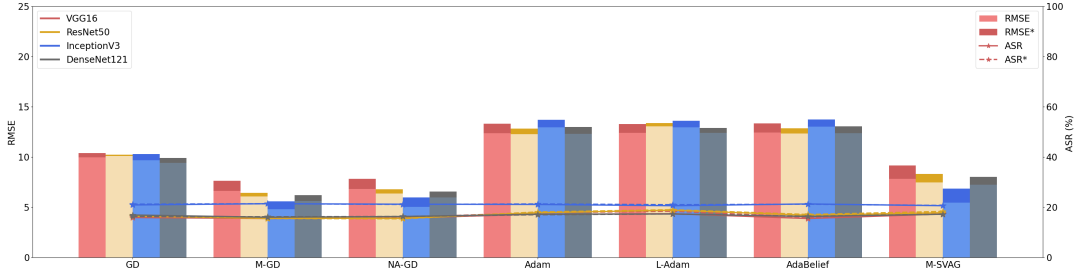


Figure 32: Transferability on ResNetXt101denoise and RMSE of the AEs with modified loss.

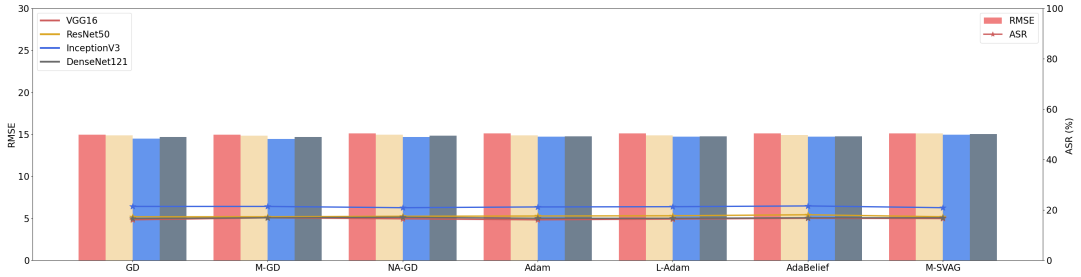


Figure 33: Transferability on ResNetXt101denoise and RMSE of the AEs with constant RMSE.

C More Results of our Counter-Example Attack

We report more results in Table 3. The trend holds that the naive strategy increases ASR towards almost all victims. The RMSE of results in Table 1 and 3 is shown in Table 4.

Table 3: ASR of the Counter-Example Attack (✓) Compared to Baselines (×)

Surrogate / Baseline		Undefended Victim						Defended Victim		
		DN201	VGG19	RN152	IncRN	Xcpt	NASL	A1	A2	A3
VGG16	×	80.7	99.1	61.3	63.7	73.1	66.4	45.0	38.1	15.7
PGD	✓	82.5	99.4	64.3	65.8	74.9	69.0	46.4	39.2	15.7
VGG16	×	86.9	99.7	73.6	76.2	82.5	73.3	57.2	52.8	18.8
SI-AoA	✓	87.9	100.0	79.9	81.4	85.5	81.4	65.0	57.6	15.9
RN50	×	65.2	52.3	46.5	34.8	35.9	25.6	19.8	15.9	15.4
PGD	✓	70.2	56.4	50.0	37.6	39.6	29.1	20.3	16.9	15.7
RN50	×	75.4	66.7	65.3	48.1	47.0	35.7	28.2	23.7	15.9
SI-AoA	✓	80.4	74.4	66.5	51.9	51.8	40.7	30.2	27.8	16.8
IncV3	×	25.1	37.0	26.8	23.8	34.2	20.5	16.4	13.9	21.2
PGD	✓	28.0	41.4	27.8	25.7	40.6	24.0	17.3	15.4	20.9
IncV3	×	32.6	43.4	31.0	29.6	44.9	29.1	21.6	17.7	19.8
SI-AoA	✓	42.2	49.7	40.7	35.4	52.7	39.0	26.6	24.1	22.0
DN121	×	81.6	54.0	40.8	32.0	38.2	29.7	17.3	15.9	16.2
PGD	✓	89.1	62.8	45.6	38.4	48.4	38.0	20.4	17.5	16.4
DN121	×	89.5	66.0	53.6	43.6	47.5	41.7	30.0	25.2	18.1
SI-AoA	✓	94.8	75.8	61.9	52.5	57.9	52.2	28.9	25.6	14.4

Table 4: RMSE of the Counter-Example Attack (indicated by *) Compared to Baselines

Surrogate	Attack	RMSE	Attack	RMSE
VGG16	PGD	7.308	DI-TI-MI	12.08
	PGD*	8.064	DI-TI-MI*	14.64
	DI-TI	8.461	SI-AoA	8.651
	DI-TI*	11.52	SI-AoA*	12.10
RN50	PGD	6.389	DI-TI-MI	11.82
	PGD*	7.815	DI-TI-MI*	14.64
	DI-TI	7.796	SI-AoA	7.215
	DI-TI*	10.88	SI-AoA*	9.671
IncV3	PGD	6.174	DI-TI-MI	11.68
	PGD*	9.200	DI-TI-MI*	14.27
	DI-TI	6.786	SI-AoA	6.823
	DI-TI*	10.04	SI-AoA*	10.34
DN121	PGD	6.403	DI-TI-MI	12.09
	PGD*	8.906	DI-TI-MI*	14.59
	DI-TI	7.497	SI-AoA	7.214
	DI-TI*	10.64	SI-AoA*	11.01